

Additional HVarPart sensitivity analysis

Additional analysis and correction

During additional quality-control checks of the hierarchical variation partitioning results, we identified an error in our previous post-processing: negative adjusted contributions were not retained. This issue was identified by the authors rather than raised by the reviewers. We replaced the previous pathway with a single extractor that preserves HVarPart's `Unique`, `Average.share`, `Individual`, `I.perc(%)`, and total adjusted R^2 values.

We compared three reporting profiles:

1. **Signed:** retain each unmodified individual contribution.
2. **Zero-truncated:** replace negative individual contributions with exactly zero and recompute the denominator from the truncated contributions.
3. **Exclude-negative:** exclude the complete model if either predictor has a negative individual contribution.

Within reporting group g , the untruncated signed hierarchical contribution of predictor p is

$$A_{p,g} = \frac{\sum_m I_{p,m}}{\sum_m R_{\text{adj},m}^2},$$

where $I_{p,m}$ is HVarPart's unmodified individual contribution. Models are otherwise excluded only when required predictors are absent, the denominator is non-finite or non-positive, or an individual contribution is non-finite. Every exclusion is recorded in `Outputs/Tables/Diagnostics/HVarPart/hvarpart__model_audit.csv`.

A negative signed hierarchical contribution is an adjusted partitioning result. It is not evidence of a negative ecological effect.

Results and presentation

For spatial SPD models, the zero-truncated and renormalised shares underlying the main balance figure are 75.6% for climate and 24.4% for human impact. The corresponding untruncated signed hierarchical contributions are 76.4% and 23.6%. Excluding complete models with any negative contribution gives 72.4% for climate.

Across all five analyses, climate remains larger than human impact under all three profiles: yes.

Because the profiles support the same overall interpretation, the main figures use zero-truncated and renormalised shares of positive hierarchical contribution. Figure 2 presents the human-minus-climate balance spatially, Figure 3 shows the human, climate, and spatial shares through time, and Figure 4 uses the human-minus-climate balance within its importance insets. Signed versions of Figures 2–4 are provided in the Supplementary Information, and the exclude-negative analysis is reported as an additional sensitivity profile.

Table 1: Model-level HVarPart eligibility audit. Counts are not duplicated across predictor rows.

Analysis	Models	Eligible	Excluded	Neg. unique	Neg. individual
h2 spd	31	31	0	1	0
spatial events	1262	257	1005	75	34
spatial spd	1262	1118	144	481	265
temporal events	85	76	9	13	10
temporal spd	70	69	1	14	6

Figure changes

Figure 2 now expresses the complementary zero-truncated human and climate shares as human minus climate importance. The bounded scale runs from -1 (climate impact) through 0 (equal importance) to 1 (human impact). For each continent, the figure combines the continent-wide distribution of record-level balances, pooled climate-zone balances, and a horizontal line for the pooled continental balance. Densities use a continuous climate-to-neutral-to-human colour scale; the climate-zone panels use the same scale as a pale background while their summary points retain categorical climate-zone colours. Maps and continental density panels do not use the background gradient. The supplementary counterpart reports the untruncated human-minus-climate hierarchical contribution difference with the same spatially filtered summaries.

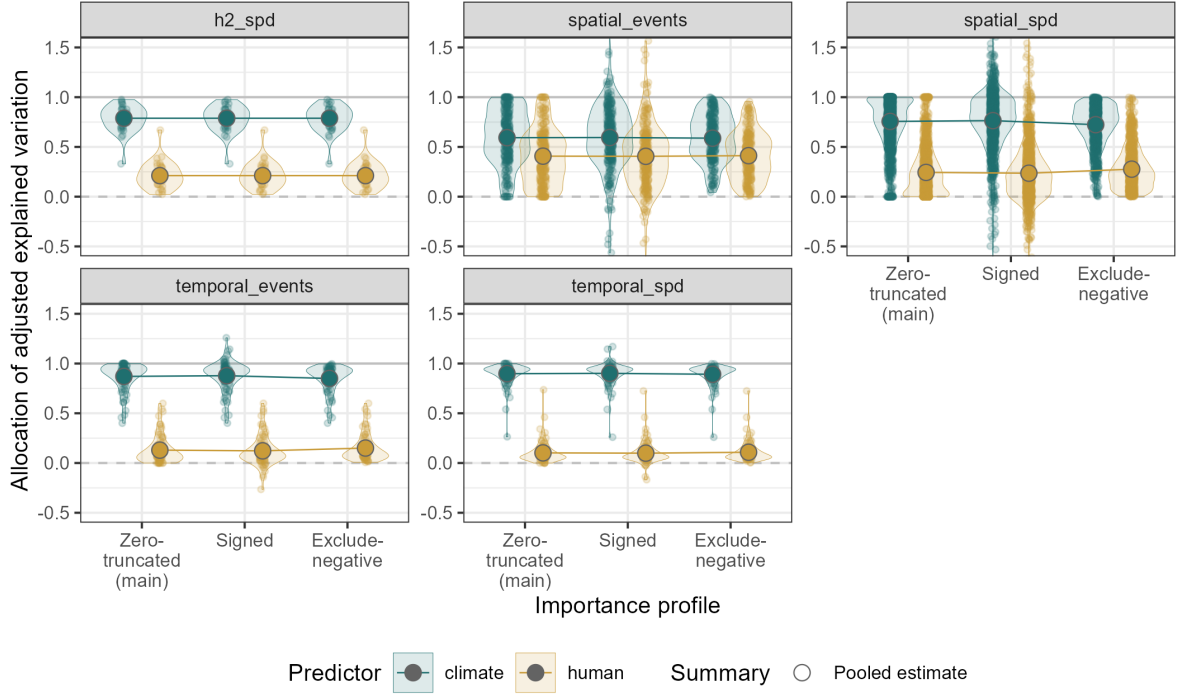
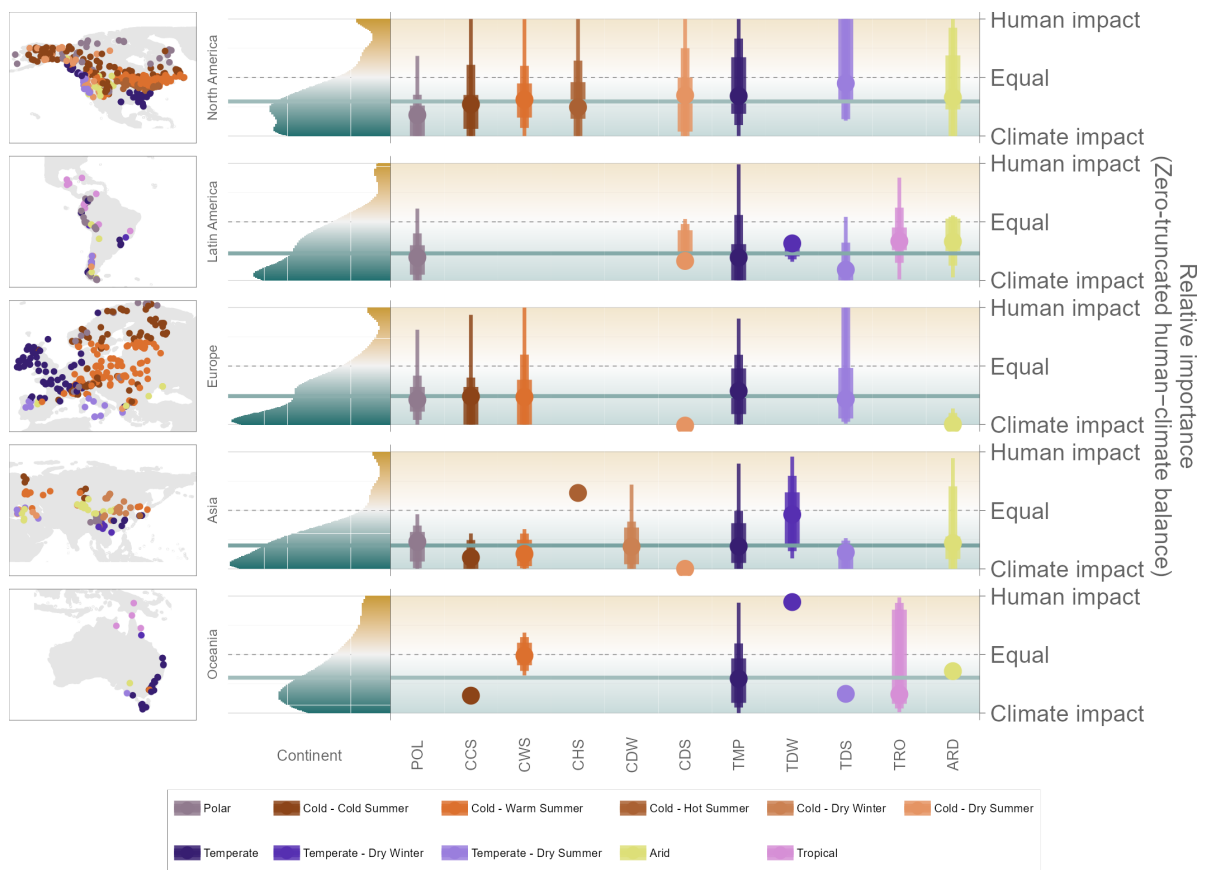
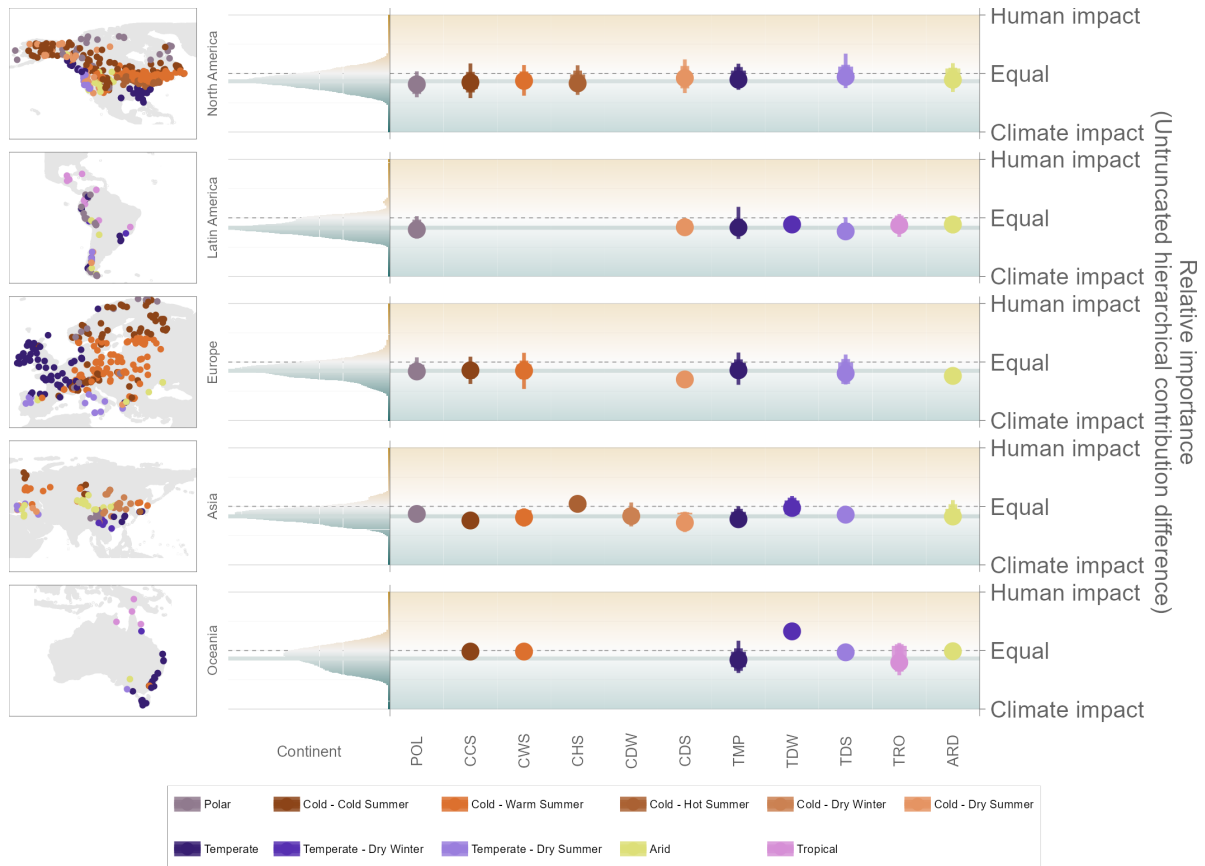


Figure 1: Zero-truncated and renormalised main-figure composition and two sensitivity profiles, calculated from the preserved fitted HVarPart results.

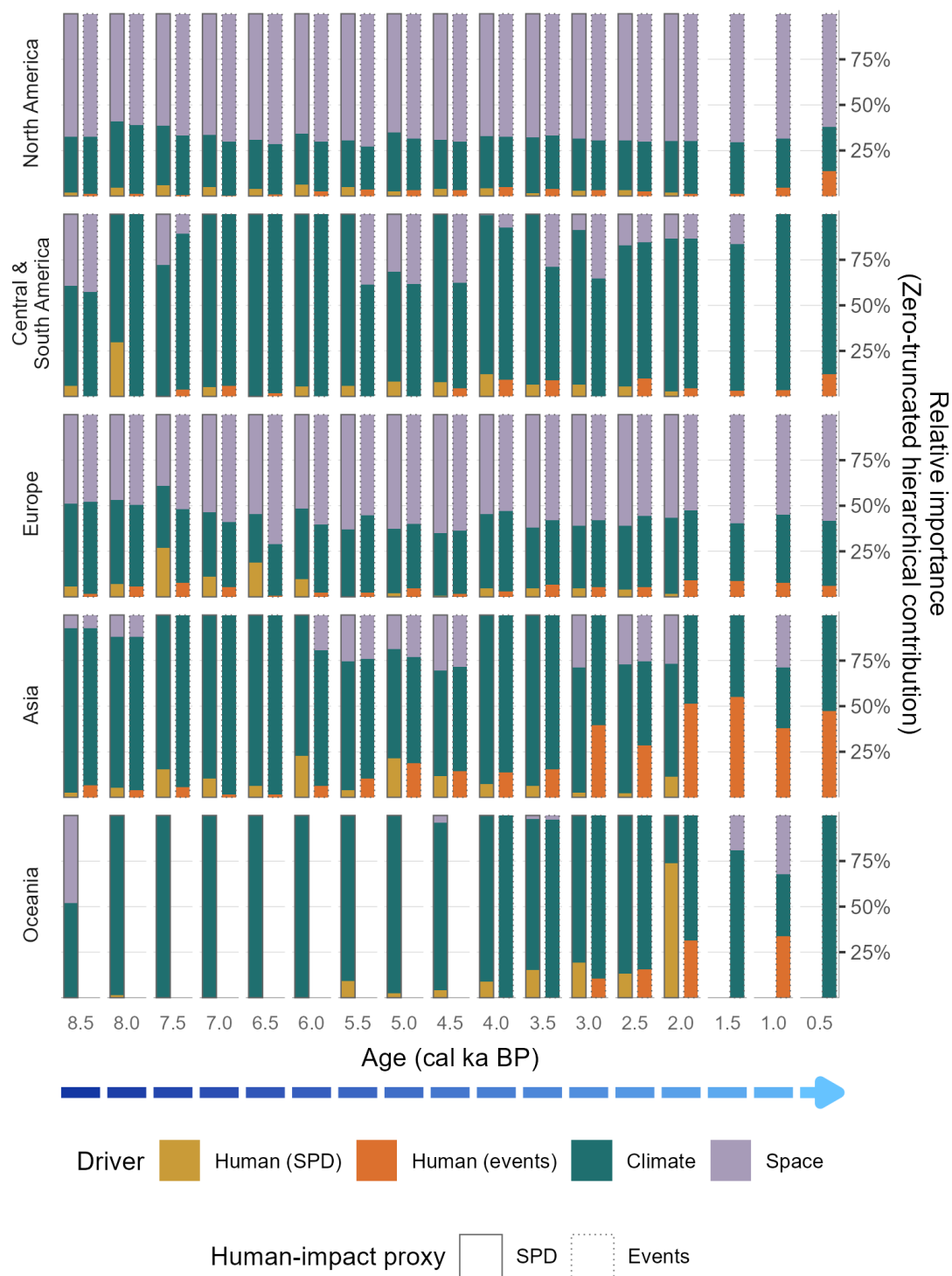


Revised main manuscript Figure 2. Spatial balance between zero-truncated and renormalised human and climate shares.

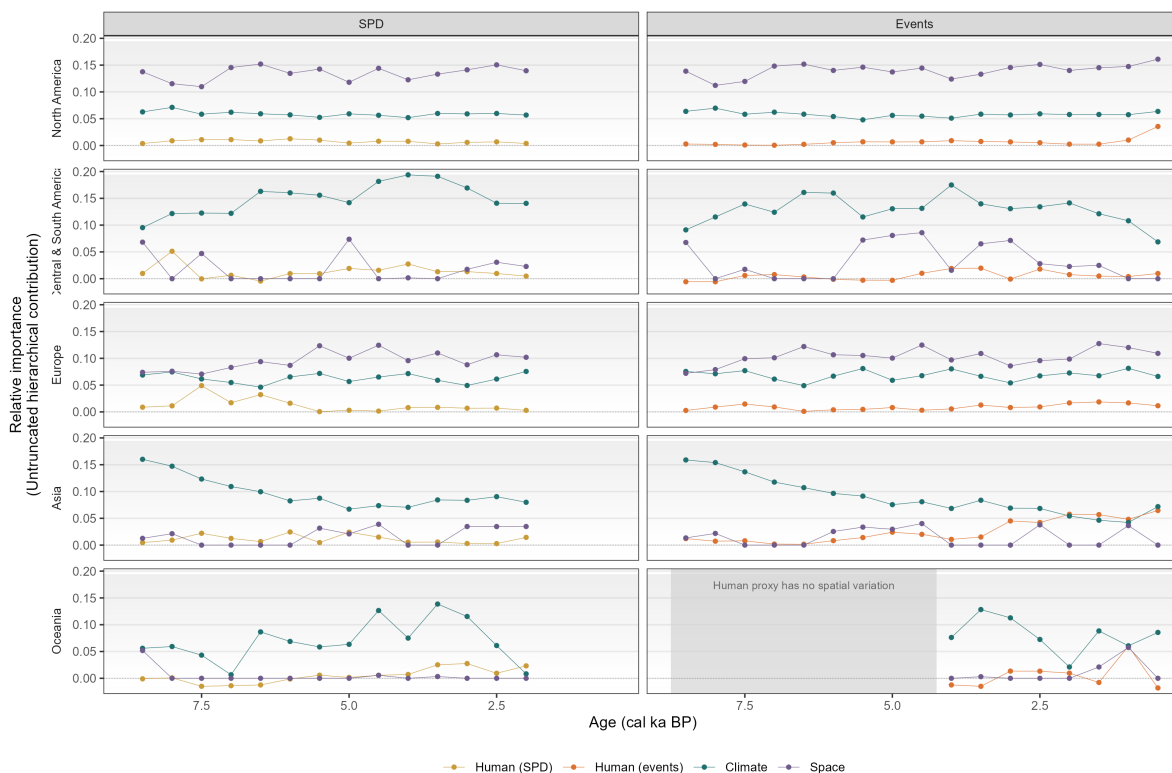


Supplementary signed Figure 2. Untruncated human-minus-climate hierarchical contribution difference with time and space control.

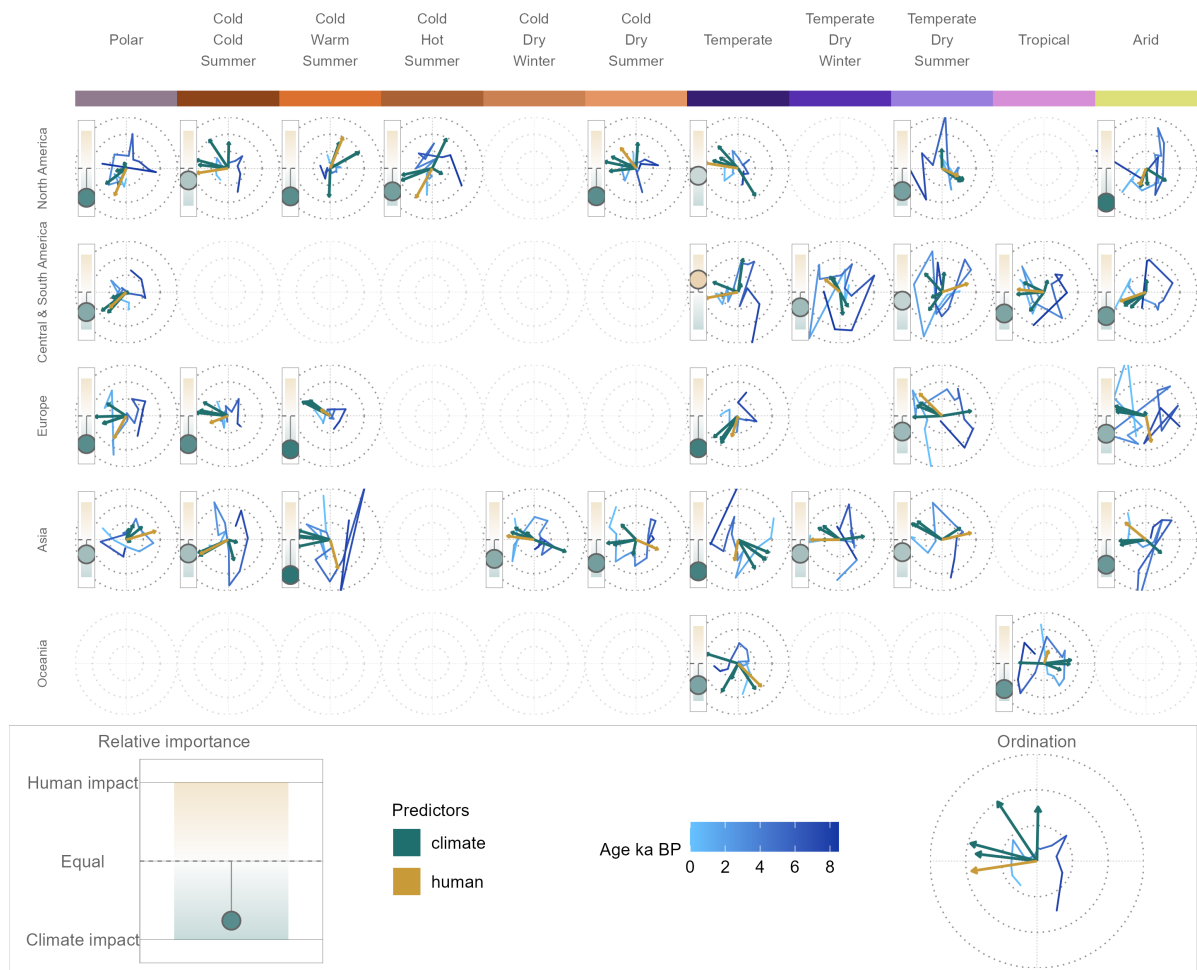
The main temporal H1 figure shows the zero-truncated and renormalised composition of positive human, climate, and spatial hierarchical contributions. Each eligible stacked bar sums to one. Gold denotes the SPD human proxy, orange denotes the event-based human proxy, teal denotes climate, and purple denotes the spatial control; bar outlines distinguish the two human proxies. SPD estimates are restricted to 2–8.5 ka BP, whereas event-based estimates span 0.5–8.5 ka BP, and the blue arrow points from older to younger ages. Its supplementary counterpart applies the same age domains and shows the untruncated signed hierarchical contributions as grouped, unstacked bars with zero and one reference lines. The H2 main-figure insets use one outlined human-minus-climate balance lollipop per region and climate zone against the pale gradient; supplementary insets retain separate untruncated signed human and climate hierarchical contributions.



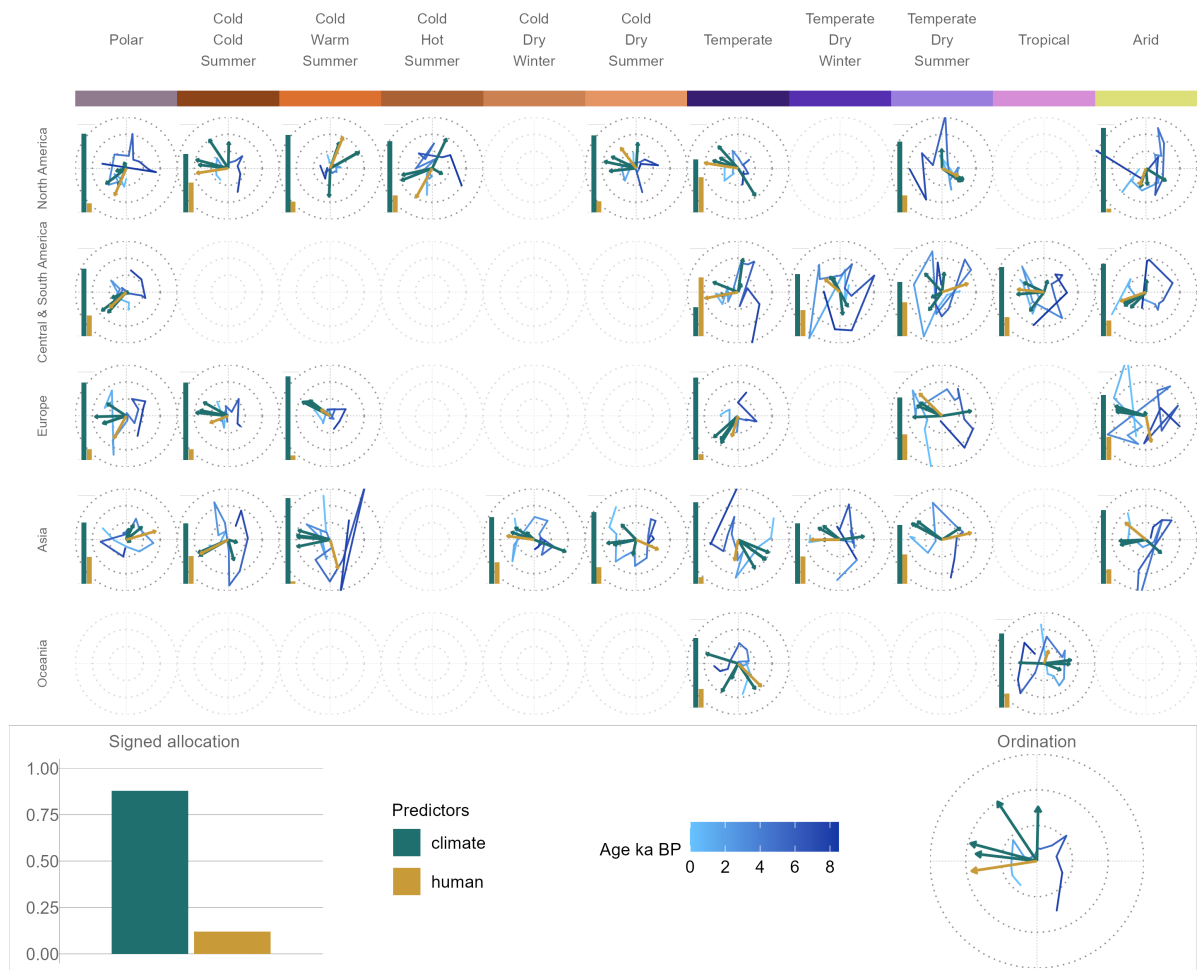
Revised main manuscript Figure 3. Temporal composition of zero-truncated and renormalised human and climate shares.



Supplementary signed Figure 3. Untruncated signed climate and human hierarchical contributions through time.



Revised main manuscript Figure 4. Multidimensional community-change ordinations with balance lollipops for human and climate importance.



Supplementary signed Figure 4. Multidimensional community-change ordinations with separate untruncated signed human and climate hierarchical contributions.